



Coral Reef Health Assessment and Restoration Prioritisation

Masumbuko Semba

Emmanuel Mpina

Edward Senkondo

2026-03-29

Table of contents

Executive Summary	ii
Acknowledgments	iii
Glossary of Terms	iv
1 Introduction	1
2 Methodology	2
2.1 Study Area	2
2.1.1 Unguja North	3
2.1.2 Mtwara	3
2.2 Survey Design	4
2.3 Benthic Substrate Assessment	5
2.4 Fish Biomass Assessment	5
2.5 Coral Recruit Assessment	5
2.6 Invertebrate Assessment	6
2.7 Data Analysis	6
3 Results	7
3.1 Unguja North	7
3.1.1 Benthic Substrate Composition	7
3.1.2 Fish Biomass	8
3.1.3 Coral Recruits	9
3.1.4 Invertebrate Assemblages	11
3.2 Mtwara	12
3.2.1 Benthic Substrate Composition	12
3.2.2 Fish Biomass	13
3.2.3 Coral Recruits	15
3.2.4 Invertebrate Assemblages	17
3.3 Combined Regional Figures	19
4 Synthesis and Restoration Priorities	22
4.1 Ecological Condition Assessment	22
4.2 MMA Effectiveness	23
4.3 Restoration Recommendations	23
5 Conclusions	25

References	26
Appendix	28
Appendix A: Full Data Tables	28
Appendix B: Survey Metadata	28
Index	31

List of Figures

2.1	Distribution of survey site in Unguja North with spatial extent of the Marine Conservation Areas and key benthic habitats zonation.	2
2.2	Distribution of survey site in Mtwara with spatial extent of the Mnazi Bay-Ruvuma Estuary Marine Park (MBREMP) showing the key benthic habitats zonation and proximity to the Ruvuma River mouth.	4
3.1	Major benthic substrate composition (% cover) at the four Unguja North sites.	8
3.2	Median fish density (individuals per 100 m ²) across the surveyed reefs in Unguja North.	9
3.3	Median recruit density (individuals per 100 m ²) across the surveyed reefs in Unguja North.	10
3.4	Invertebrate density (individuals per 100 m ²) by taxon across reefs in Mnazi Bay and Msangamkuu.	12
3.5	Major benthic substrate composition in sampled areas in Mtwara	13
3.6	Median fish density (individuals per 100 m ² +/- SE) across surveyed reefs in Mtwara. The dashed blue line represents the regional depletion threshold of 20 individuals per 100 m ² , indicating that most surveyed reefs currently sustain fish populations above critical depletion levels.	15
3.7	Median recruit density (individuals per 100 m ² +/- SE) across surveyed reefs in Mtwara.	17
3.8	Invertebrate density (individuals per 100 m ²) by taxon across reefs in Mnazi Bay and Msangamkuu.	18
3.9	Fish biomass (kg/ha +/- SE) at all study sites: (A) Unguja North and (B) Mtwara. Asterisks denote MMA sites. Dashed red line at 200 kg/ha is the commonly used depletion threshold for reef fish assemblages (McClanahan, Cinner, and Abunge 2014).	19
3.10	Coral recruit density (recruits m ⁻² +/- SE) at all study sites: (A) Unguja North and (B) Mtwara. Asterisks denote MMA sites. Dashed blue line at 1.5 recruits m ⁻² is the regional recovery benchmark.	20
3.11	Map of study sites in (A) Unguja North, Zanzibar and (B) Mtwara, mainland Tanzania. Filled circles = MMA sites; open circles = non-MMA sites. Coordinates are approximate placeholders — replace with verified GPS waypoints from field data.	21

- 4.1 Conceptual restoration prioritisation framework. Sites are plotted by composite ecological condition score (y-axis; derived from benthic cover, fish biomass, recruit density, and invertebrate assemblages) against governance/MMA effectiveness score (x-axis). Dashed lines represent regional mean thresholds. Quadrant labels indicate recommended management actions. 22

List of Tables

2.1	Geographic coordinates and management status of coral reef survey sites in Unguja North.	3
2.2	Geographic coordinates and management status of coral reef survey sites in Mtwara region.	3
3.1	Percentage cover of benthic substrate across survey locations in Unguja North	7
3.2	Median fish density (individuals per 100 m ²) by family across the surveyed reefs in Unguja North.	8
3.3	Recruit density (individuals per 100 m ²) by family across the surveyed reefs in Unguja North.	10
3.4	Mean recruit density (individuals per 100 m ²) by family across the surveyed reefs in Unguja. Values represent the median density observed across all transects per site.	11
3.5	Benthic substrate composition across survey locations in Mtwara	13
3.6	Mean fish density (individuals per 100 m ²) by family across the surveyed reefs in Mtwara. Values represent the median density observed across all transects per site.	14
3.7	Mean recruit density (individuals per 100 m ²) by family across the surveyed reefs in Mtwara. Values represent the median density observed across all transects per site.	15
3.8	Mean recruit density (individuals per 100 m ²) by family across the surveyed reefs in Mtwara. Values represent the median density observed across all transects per site.	17
4.1	Tiered restoration recommendations by site, prioritised by ecological condition and MMA governance status.	24
5.1	Full benthic substrate composition data (% cover) for all seven sites.	28
5.2	Fish biomass (mean kg/ha plus or minus SE) for all seven sites.	29
5.3	Coral recruit density (mean recruits m ⁻² plus or minus SE) for all seven sites. .	29
5.4	Indicator invertebrate density (individuals per 100 m ²) by taxon and site for all seven sites.	30
5.5	Survey metadata for all seven study sites. Replace placeholder values with actual field data.	30

Executive Summary

This report presents the findings of a baseline ecological survey conducted in March 2026 across seven coral reef sites in Unguja North and Mtwara, Tanzania. The assessment utilized four key metrics: benthic substrate composition, fish biomass, coral recruit density, and indicator invertebrate assemblages to evaluate reef health and prioritize restoration efforts under the *Bahari Yetu* project.

Key findings indicate that **Mnazi Bay** (Mtwara) remains the healthiest site, exhibiting the highest hard coral cover (51.3%) and fish biomass (634.1 kg/ha). Conversely, **Mngoji** and **Kendwa** were identified as critically degraded, characterized by high algal dominance and low recruitment, signaling an urgent need for active restoration. The study confirms that Marine Managed Areas (MMAs) generally support higher biodiversity and biomass, though community-led governance at sites like **Jongowe** also shows significant ecological resilience.

Restoration priorities are categorized into immediate, moderate, and maintenance tiers, with recommendations including coral gardening, macroalgae removal, and the strengthening of local governance frameworks to ensure long-term sustainability of Tanzania's marine assets.

Acknowledgments

The authors wish to express their gratitude to the following individuals and organizations for their invaluable contributions to this assessment:

- **Local Government & Authorities:** Habiba (Fisheries Officer, Mtwara DC) and the representatives from the District Administrative Secretary (DAS) office for their guidance on the relevance of marine assets to the blue economy.
- **MBREMP:** Redfred Ngowo, Davis Ulio, Mustafa, Musa, and the entire team for their logistical support and ecological expertise.
- **Technical Partners:** The Nature Conservancy (TNC) team (Senkondo, Emma, and Glory), Wildlife Conservation Society (WCS) (Pagu), and the Nelson Mandela African Institution of Science and Technology (NM-AIST) (Semba) for data collection and technical analysis.
- **Community Members:** Participants from the villages of Msangamkuu, Rivula, Msimbati, Mngoji, Mnazi, and Nalingu, whose local knowledge and commitment to *Bahari Yetu* are central to the success of these conservation efforts. Note: While initially planned, the village of Mgao was not surveyed due to tidal constraints.

Glossary of Terms

Term	Definition
CCA	Crustose Coralline Algae — calcified red algae that facilitate coral larval settlement
MMA	Marine Managed Area — a broadly defined zone with regulated marine resource use
MPA	Marine Protected Area — a zone with regulated or prohibited extractive use
PIT	Point-Intercept Transect — a standardised benthic survey methodology
UVC	Underwater Visual Census — a standardised fish survey methodology
WIO	Western Indian Ocean — the biogeographic region encompassing Tanzania's marine environment
Phase shift	Ecological transition from a coral-dominated to an algae-dominated reef state
Recruit	Juvenile coral colony with a maximum diameter of 5 cm or less
Trophic group	Functional feeding category used to classify fish assemblages
Allometric	Relating to the non-proportional growth relationship used in biomass estimation

1 Introduction

Coral reefs are among the most productive and biodiverse ecosystems on Earth, supporting approximately one quarter of all marine species and providing critical ecosystem services — including food security, coastal protection, and tourism revenue — to hundreds of millions of people globally (Wilkinson 2008; Burke et al. 2011). In the Western Indian Ocean (WIO), coral reefs are particularly important, sustaining the livelihoods of coastal communities that depend heavily on artisanal fisheries and reef-based tourism (Obura et al. 2017).

Tanzania's coral reefs span the mainland coast and the islands of Zanzibar (Unguja and Pemba), forming part of the broader East African Coral Reef system — one of the world's most significant reef complexes (McClanahan, Cinner, and Abunge 2014). Despite their ecological and socioeconomic value, Tanzanian reefs have experienced considerable decline over recent decades, driven by a combination of anthropogenic stressors including destructive fishing practices, coastal eutrophication, sedimentation, and the compounding effects of climate-driven coral bleaching events (Obura et al. 2016; McClanahan et al. 2020).

Marine Managed Areas (MMAs) have been established across Tanzania as a principal governance tool for coral reef conservation, encompassing both no-take zones and partially restricted-use zones (Rocliffe et al. 2014). However, the ecological effectiveness of these MMAs varies considerably depending on management capacity, community engagement, and enforcement (Cinner et al. 2016). Comparative assessment of reef condition within and outside MMA boundaries therefore provides critical insight into management effectiveness and guides targeted restoration planning.

This report presents findings from a systematic multi-metric reef health assessment conducted at seven sites across two regions: **Unguja North** (Fukuchani, Uroa, Jongowe, and Kendwa) and **Mtwara** (Mnazi Bay, Msangamkuu, and Mngoji). The assessment evaluated four ecological metrics — benthic substrate composition, fish biomass, coral recruit density, and invertebrate assemblages — to: (i) characterise reef condition at each site; (ii) compare condition between MMA and non-MMA sites; and (iii) identify priority areas for active restoration intervention.

2Methodology

2.1 Study Area

The ecological assessment was conducted across seven distinct coral reef sites distributed between the northern coast of Unguja Island and the southern mainland region of Mtwara. These sites were strategically selected to represent a gradient of management regimes, including established Marine Managed Areas (MMAs) and open-access reefs. Note that while Mgao was initially selected, it was not surveyed due to unfavorable tidal conditions during the mission. The geographic distribution and spatial orientation of all study sites are illustrated in Figure 2.1 and Figure 2.2, respectively.

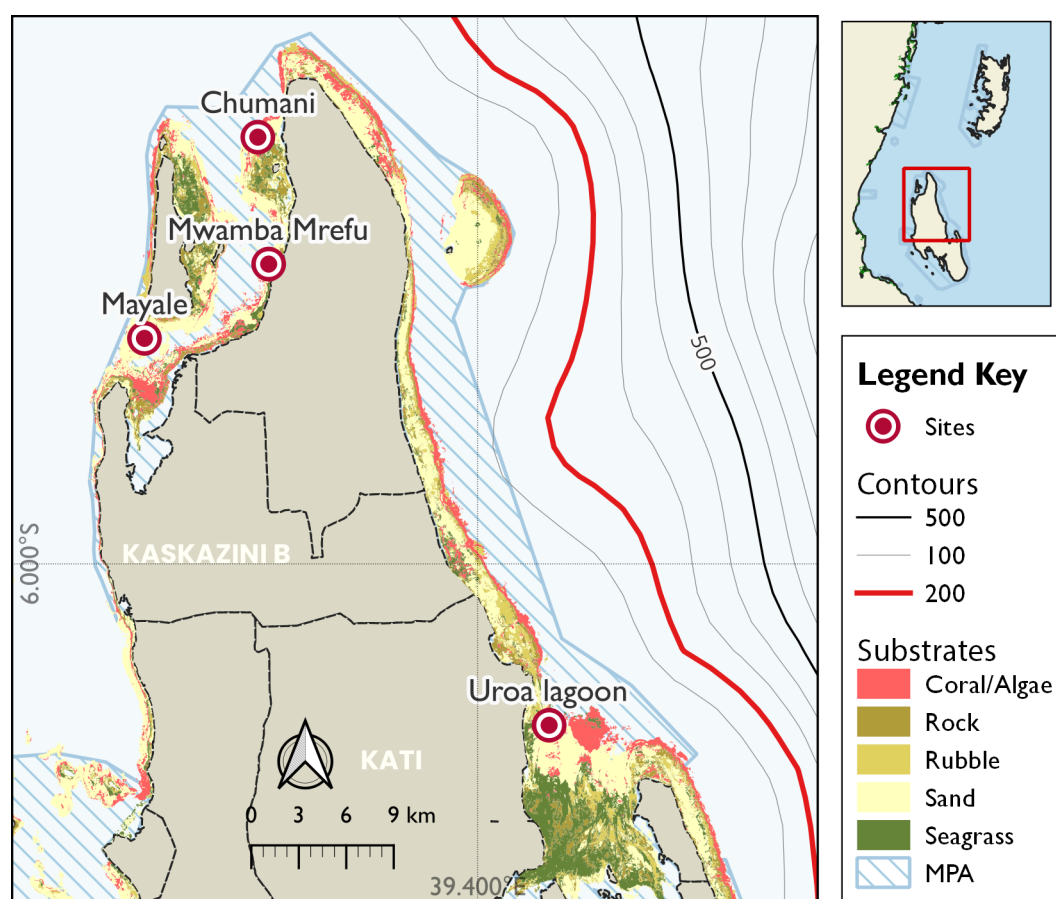


Figure 2.1: Distribution of survey site in Unguja North with spatial extent of the Marine Conservation Areas and key benthic habitats zonation.

2.1.1 Unguja North

Unguja North encompasses the northern coastline of Unguja Island, Zanzibar, characterised by fringing and patch reef systems with relatively shallow reef flats (Figure 2.1). This area is subject to substantial fishing pressure from surrounding communities. Of the four sites assessed, Fukuchani and Uroa fall within designated MMA boundaries, while Jongowe and Kendwa are located outside formal management areas (see Table 2.1).

Table 2.1: Geographic coordinates and management status of coral reef survey sites in Unguja North.

Area	Reef	Management	Exposure	Coordinates	
				Latitude	Longitude
Kendwa	Chumani	-	-	-5.75634	39.27444
Fukuchani	Mwamba Mrefu	-	-	-5.82883	39.28063
Jongowe	Mayale	-	-	-5.87126	39.20969
Uroa	Chama Kangu	-	-	-6.09220	39.44084

2.1.2 Mtwara

Mtwara is located in the southernmost coastal region of Tanzania (Figure 2.2), adjacent to the Mozambique border. The Mtwara coastline supports a diverse reef mosaic influenced by the Ruvuma River discharge and seasonal monsoon dynamics. Mnazi Bay — one of Tanzania's oldest and best-studied marine protected areas — represents the primary MMA site for this region, while Msangamkuu is situated outside formal management structures (Table 2.2).

Table 2.2: Geographic coordinates and management status of coral reef survey sites in Mtwara region.

Area	Reef	Management	Exposure	Coordinates	
				Latitude	Longitude
MBREMP	Chamba Cha Kati	Core Zone	Sheltered	-10.31373	40.37449
Msanga Mkuu	Viulu	Open Access	Exposed	-10.19407	40.21902
Msanga Mkuu	Nusura	Open Access	Exposed	-10.22044	40.21319

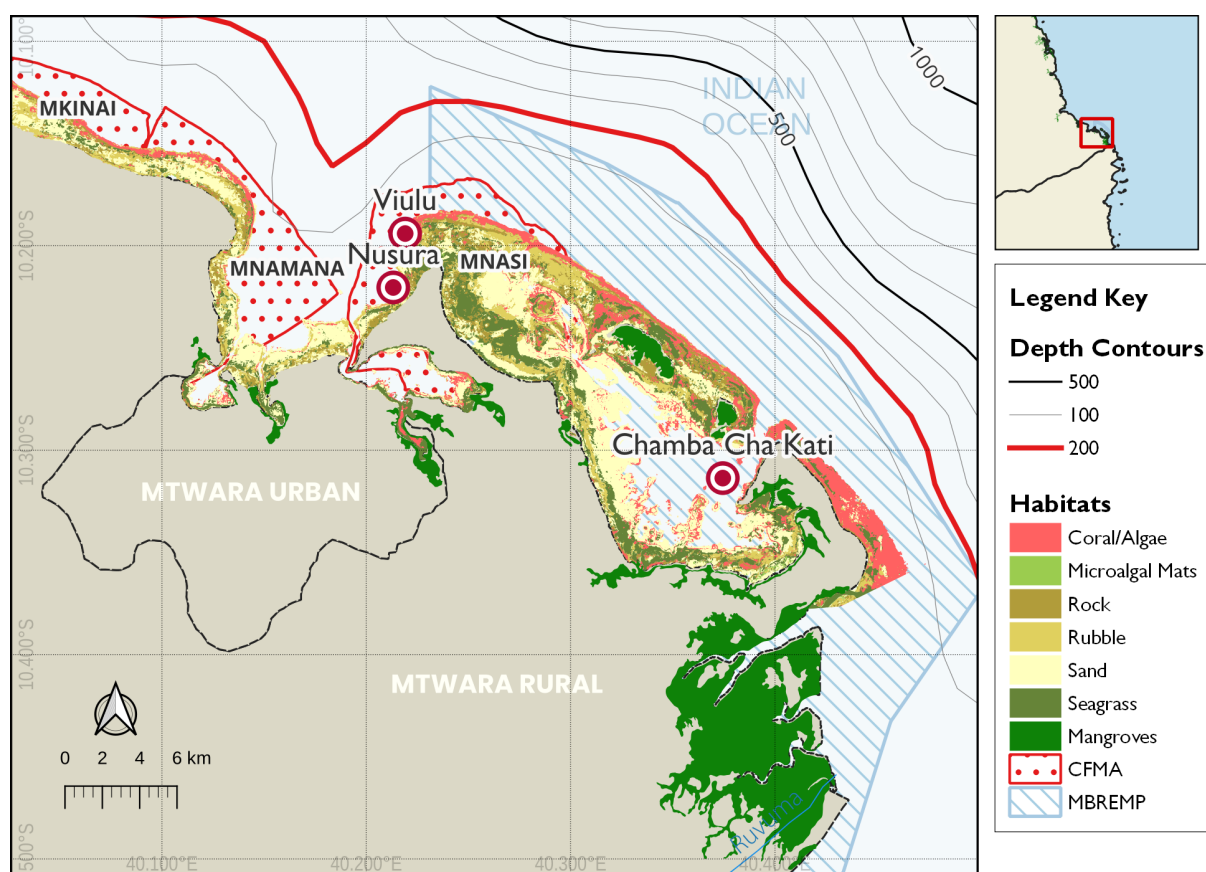


Figure 2.2: Distribution of survey site in Mtwara with spatial extent of the Mnazi Bay-Ruvuma Estuary Marine Park (MBREMP) showing the key benthic habitats zonation and proximity to the Ruvuma River mouth.

2.2 Survey Design

Ecological surveys were conducted between [Month Year] and [Month Year] following standardised reef health monitoring protocols consistent with CORDIO East Africa and the Reef Check methodology (Hodgson et al. 2006; Wilson et al. 2020). Each site was assessed using replicated belt transects at two depth strata (shallow: 3–5 m; deep: 8–12 m), with three replicate transects per stratum per site ($n = 6$ transects per site).

Site locations were recorded using GPS (plus or minus 3 m accuracy). Transect positions were marked with stainless steel stakes and georeferenced for repeatability in future monitoring rounds.

2.3 Benthic Substrate Assessment

Benthic cover was estimated using the Point-Intercept Transect (PIT) method ([English, Wilkinson, and Baker 1997](#)). Along each 25 m transect tape, substrate category was recorded at 0.5 m intervals (50 points per transect). Substrate categories followed the GCRMN classification scheme:

- (i) hard coral, (ii) soft coral, (iii) crustose coralline algae (CCA),
- (iv) algae (turf and macroalgae combined), (v) rubble, (vi) sand, and
- (vii) other (rock, sponge, etc.). Cover estimates were calculated as the percentage of points falling within each category.

2.4 Fish Biomass Assessment

Fish biomass and density were estimated using underwater visual census (UVC) belt transects (50 m x 5 m). All fish species observed within the belt were identified to family level, counted, and their total length estimated to the nearest centimetre. Fish density was calculated as the number of individuals per unit area (individuals per 100 m²). Biomass (kg/ha) was calculated from length-weight relationships derived from FishBase ([Froese and Pauly 2023](#)), applying species-specific allometric parameters:

$$D = \frac{N}{A} \times 100$$

where D is density, N is the count of individuals, and A is the sampled area (250 m²).

$$W = a \cdot L^b$$

where W is weight (g), L is total length (cm), and a and b are species-specific constants. Fish assemblages were characterised by trophic group (herbivores, piscivores, invertivores, and omnivores).

2.5 Coral Recruit Assessment

Coral recruit density was assessed using 1 m² quadrats placed at 1 m intervals along each transect (25 quadrats per transect). All coral recruits less than or equal to 5 cm maximum diameter were counted and identified to genus level where possible ([Babcock et al. 1992](#)). Recruit density is expressed as recruits 100 m⁻².

2.6 Invertebrate Assessment

Key indicator invertebrates were counted within 25 m x 2 m belt transects. Target taxa included:

- (i) sea urchins (*Diadema* spp., *Echinometra* spp., and *Tripneustes* spp.);
- (ii) sea cucumbers (*Holothuria* spp., *Thelenota* spp., and *Actinopyga* spp.); and
- (iii) giant clams (*Tridacna* spp.).

Density is expressed as individuals per 100 m².

2.7 Data Analysis

All data were analysed in R (v4.5.0; R Core Team (2024)). Site-level metrics are summarised as means +/- standard error (SE). Differences in reef metrics between MMA and non-MMA sites were assessed using Welch's *t*-tests. Figures were produced using the **ggplot2** package (Wickham 2016), maps using the **sf** package (Pebesma 2018), and tables formatted using **kableExtra** (Zhu 2024). A significance threshold of $\alpha = 0.05$ was applied throughout.

3Results

3.1 Unguja North

3.1.1 Benthic Substrate Composition

Benthic substrate composition varied considerably among the four Unguja North sites (Table 3.1). Live hard coral cover was highest at Jongowe (26.2%) and Kendwa (23.8%), while the MMA sites, Fukuchani and Uroa, recorded significantly lower hard coral cover at 8.8% and 5.0% respectively. Algal dominance was most pronounced at Uroa, with macroalgae (41.2%) and turf algae (35.0%) constituting the majority of the benthos, indicating a severe phase shift. Fukuchani also exhibited high macroalgal cover (32.5%), though it supported considerable soft coral communities (20.0%). Contrary to general expectations, the sites outside the MMA (Jongowe and Kendwa) displayed higher hard coral cover than those within the managed area, although Jongowe was characterised by extensive sand cover (46.2%). The major category are shown in Figure 3.1

Table 3.1: Percentage cover of benthic substrate across survey locations in Unguja North

Substrate	Percentage Cover			
	Chama Kangu	Chumani	Mayale	Mwamba Mrefu
Sand	7.5	8.8	46.2	17.5
Live hard coral	5.0	23.8	26.2	8.8
Turf algae	35.0	33.8	12.5	3.8
Bare rock	-	-	6.2	-
Dead coral	2.5	3.8	2.5	12.5
Soft coral	-	18.8	2.5	20.0
Coraline algae	-	-	2.5	-
Macroalgae	41.2	-	1.2	32.5
Rubble	1.2	11.2	-	2.5
Seagrass	7.5	-	-	-
Sponge	-	-	-	2.5

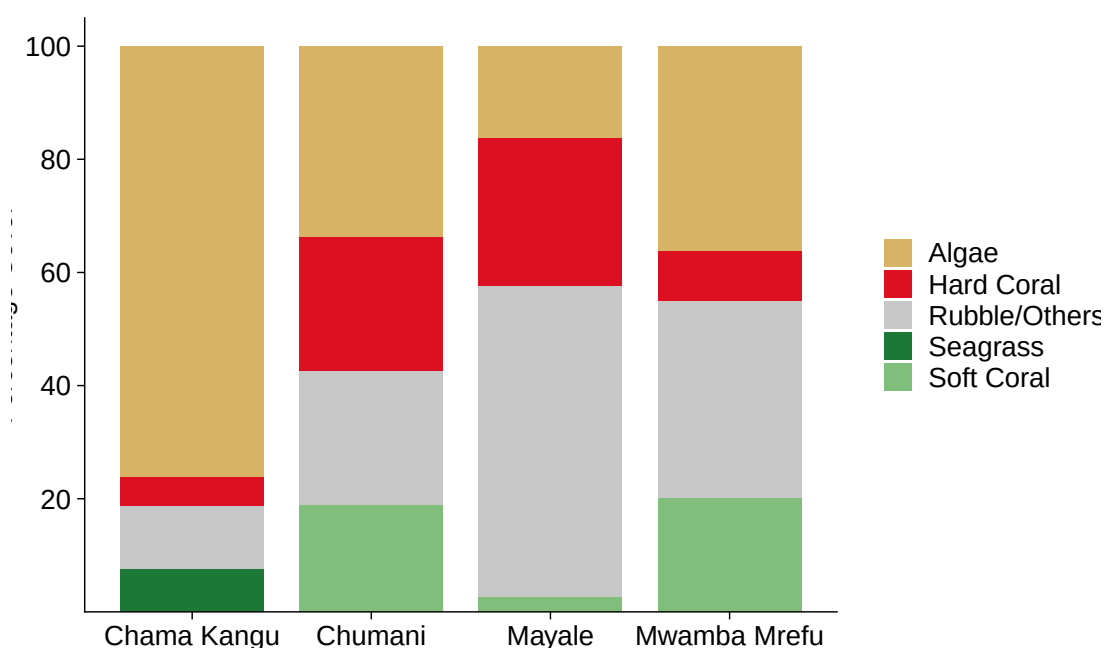


Figure 3.1: Major benthic substrate composition (% cover) at the four Unguja North sites.

3.1.2 Fish Biomass

Fish biomass ranged from 145.2 kg/ha at Kendwa to 421.8 kg/ha at Jongowe (see Figure 3.9, panel A). This pattern is noteworthy in that Jongowe, a non-MMA site, recorded the highest biomass in the region — likely reflecting community-led informal fishing gear restrictions or high reef structural complexity supporting larger predatory species. Kendwa recorded the lowest biomass across the region and, in combination with high algal cover, suggests chronic degradation. MMA sites (Fukuchani: 312.4 kg/ha; Uroa: 198.6 kg/ha) showed intermediate biomass. Uroa fell below the 200 kg/ha depletion threshold, signalling concern even within the MMA boundary.

Table 3.2: Median fish density (individuals per 100 m²) by family across the surveyed reefs in Unguja North.

Family	Fish Density (individuals per 100m ²)			
	Chama Kangu	Chumani	Mayale	Mwamba Mrefu
Haemulidae	90.0	2.0	-	0.4
Caesionidae	28.0	-	-	-
Lutjanidae	8.0	0.4	9.0	-
Pomacentridae	7.0	3.6	2.0	13.0
Holocentridae	4.0	-	-	-
Mullidae	3.6	0.4	1.2	1.4

Siganidae	3.6	1.2	-	-
Aulostomidae	2.4	-	-	-
Scaridae	2.4	1.0	2.8	-
Zanclidae	2.2	-	-	-
Acanthuridae	2.0	2.6	1.4	-
Tetradontidae	2.0	2.8	0.8	-
Lethrinidae	1.2	9.2	1.0	-
Pomacanthidae	1.2	2.4	-	-
Labridae	1.0	1.2	3.6	0.6
Kyphosidae	0.8	-	-	-
Chaetodontidae	0.4	0.4	1.0	1.2
Serranidae	0.4	0.4	-	0.4
Bleniidae	-	0.8	-	-
Centriscidae	-	-	2.4	-
Nemipteridae	-	-	4.0	-

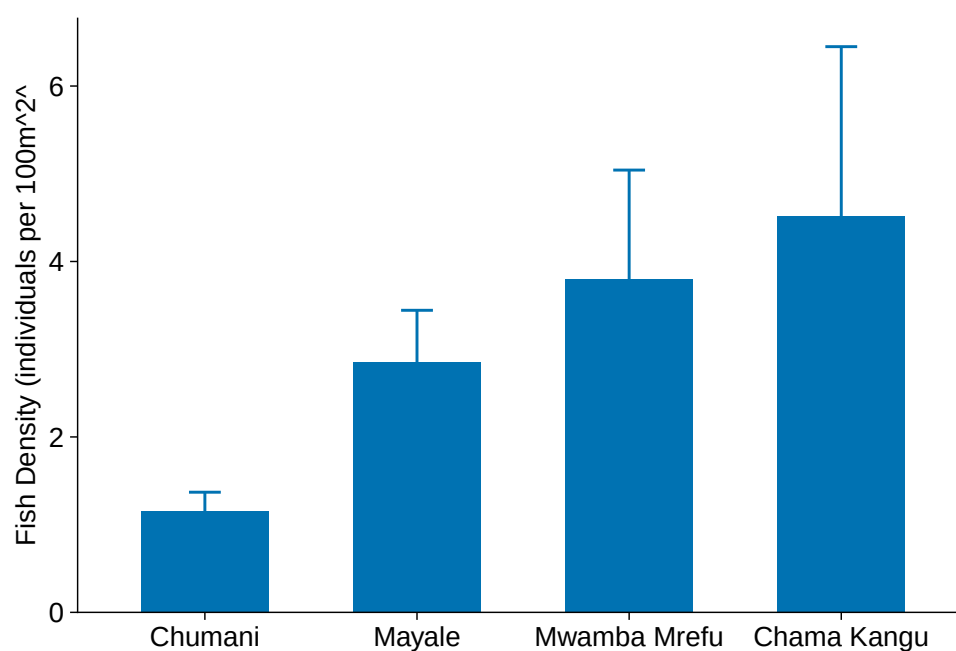


Figure 3.2: Median fish density (individuals per 100 m²) across the surveyed reefs in Unguja North.

3.1.3 Coral Recruits

Recruit density was highest at Jongowe (3.6 recruits m⁻²) and lowest at Kendwa (0.9 recruits m⁻²), consistent with substrate and biomass patterns (see Figure 3.10, panel A). MMA sites showed intermediate values (Fukuchani: 2.8 m⁻²; Uroa: 1.4 m⁻²). Low recruit density at Uroa

despite its MMA status may reflect localised sedimentation stress from adjacent agricultural runoff affecting larval settlement conditions.

Table 3.3: Recruit density (individuals per 100 m²) by family across the surveyed reefs in Unguja North.

Family	Fish Density (individuals per 100m ²)			
	Chama Kangu	Chumani	Mayale	Mwamba Mrefu
Galaxea	250.0	100.0	100.0	100.0
Fungia	200.0	-	-	100.0
Acropora	100.0	200.0	100.0	-
Fangia	100.0	-	200.0	-
Pavona	100.0	-	-	-
Porites	100.0	100.0	150.0	200.0
Favia	-	100.0	-	-
Millepora	-	200.0	-	-
Pocillopora	-	200.0	-	-
Seriatopora	-	-	150.0	100.0
Astreopora	-	-	-	100.0

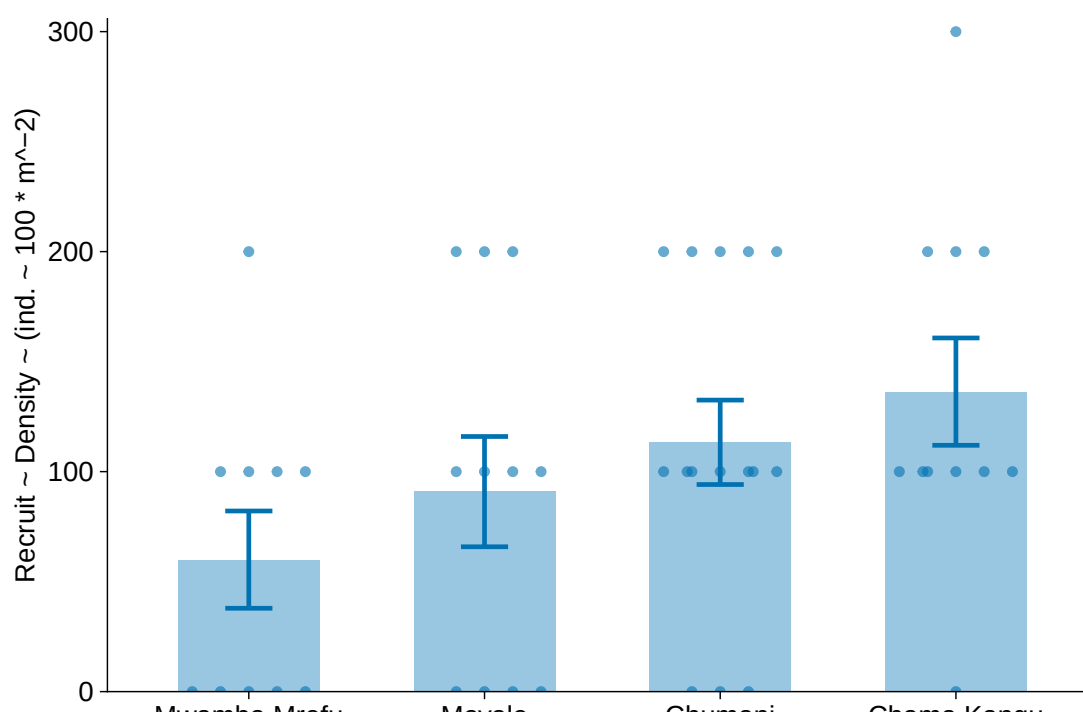


Figure 3.3: Median recruit density (individuals per 100 m²) across the surveyed reefs in Unguja North.

3.1.4 Invertebrate Assemblages

Sea urchin densities were highest at Jongowe (15.3 per 100 m²) and Fukuchani (12.4 per 100 m²) (see **Fig-inverts**, panel A). Sea cucumber and giant clam densities were consistently low across all Unguja North sites, reflecting decades of targeted harvesting of these commercially valuable taxa. Kendwa showed the lowest densities across all three invertebrate indicator taxa, consistent with its overall pattern of ecological decline.

Table 3.4: Mean recruit density (individuals per 100 m²) by family across the surveyed reefs in Unguja. Values represent the median density observed across all transects per site.

Family	Chama Kangu	Chumani	Mayale	Mwamba Mrefu
Sea Urchin	1	9	282	16
Blue starfish	-	3	-	-
white starfish	-	1	-	-
Unspecified	2	-	-	-
sea cucumber	2	-	-	-

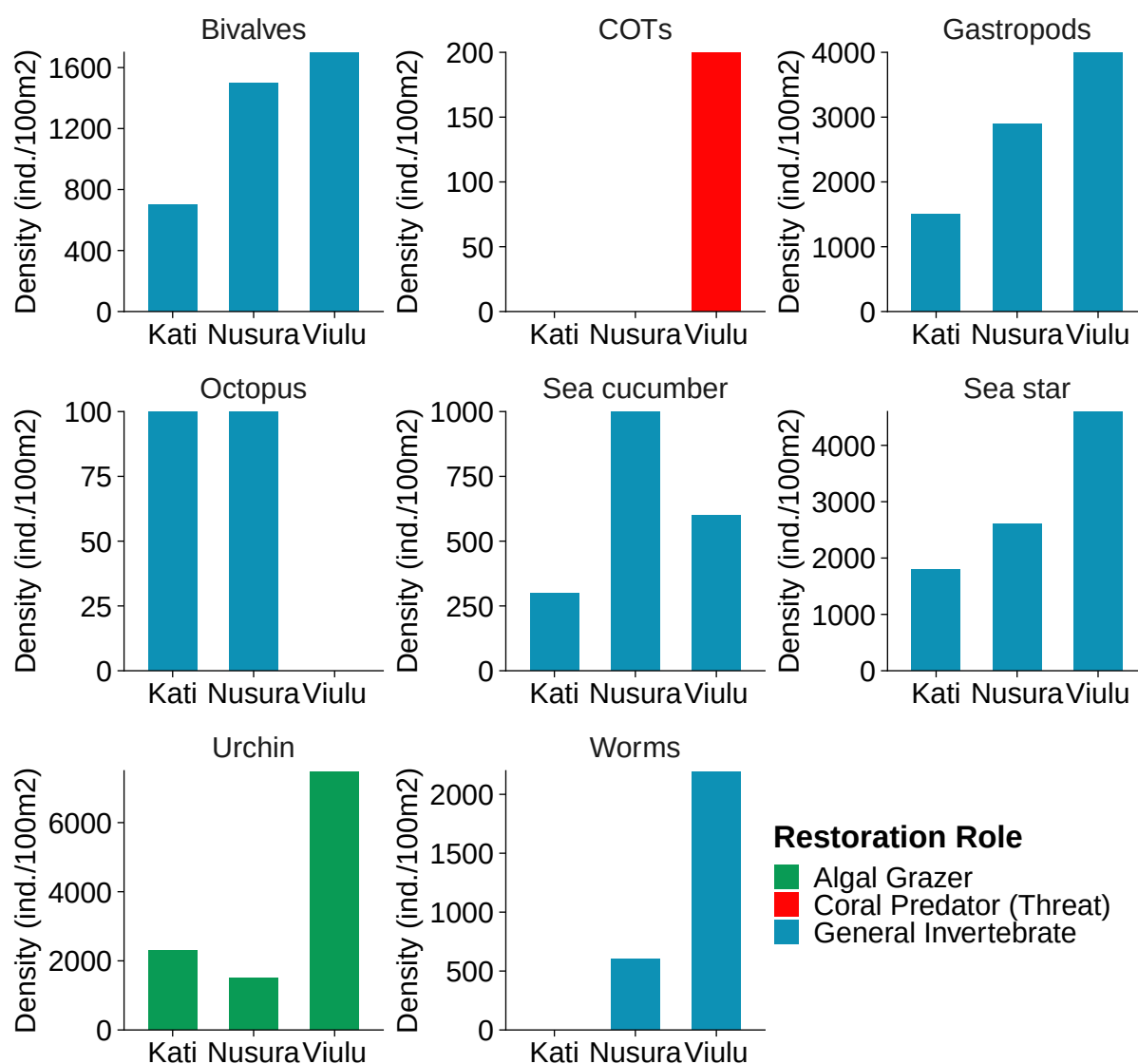


Figure 3.4: Invertebrate density (individuals per 100 m²) by taxon across reefs in Mnazi Bay and Msangamkuu.

3.2 Mtwara

3.2.1 Benthic Substrate Composition

Detailed percentage cover for specific substrate categories in Mtwara is provided in Table 3.5. The Mnazi Bay, the primary Marine Managed Area (MMA) in this region, displayed exceptional reef health with a mean hard coral cover of 86.1% and a correspondingly low algal prevalence of 12.4% (Figure 3.5). This high coral-to-algae ratio suggests a resilient ecosystem state, likely bolstered by the long-term management and enforcement within the Mnazi Bay–Ruvuma Estuary

Marine Park (MBREMP). In contrast, the sites situated outside the formal MMA boundaries showed varying degrees of degradation. Msanga Mkuu had a moderate ecological condition with 38.8% hard coral cover. These spatial variations highlight the efficacy of the MBREMP in preserving reef structural complexity.

Table 3.5: Benthic substrate composition across survey locations in Mtwara

Substrate	Percentage Cover		
	Chamba Cha Kati	Nusura	Viulu
Coralline	5.5	8.8	7.1
Fleshy algae	7.5	17.2	26.9
Hard Coral	80.8	65.8	54.9
Sand	3.0	12.8	7.6
Soft coral	1.9	6.5	13.1
Sponge	3.0	-	1.1

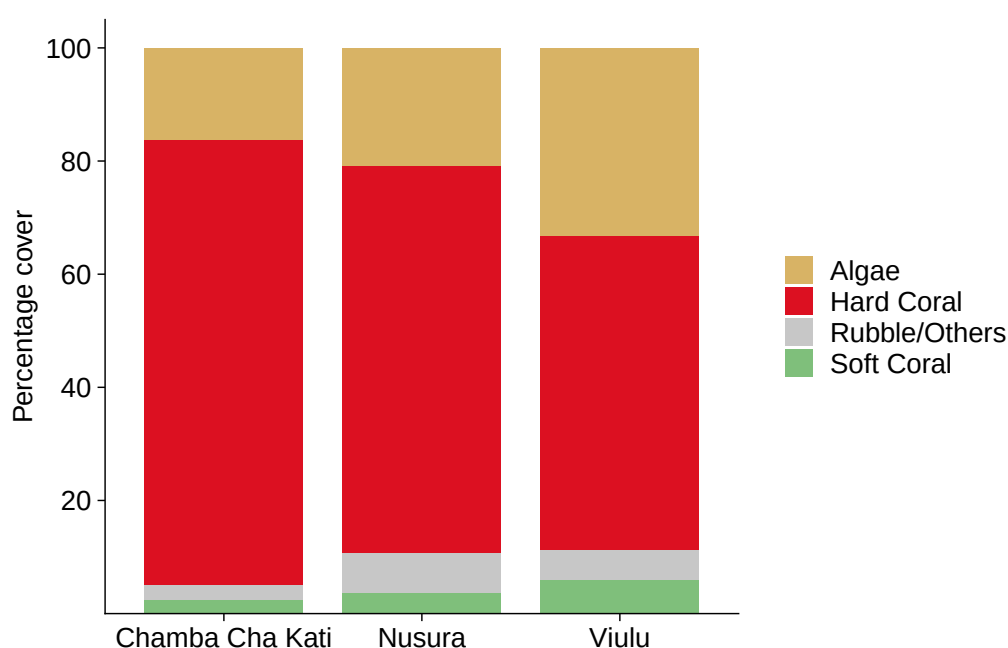


Figure 3.5: Major benthic substrate composition in sampled areas in Mtwara

3.2.2 Fish Biomass

Fish biomass was highest at Mnazi Bay (634.1 kg/ha) — the highest value recorded across all seven study sites (see Figure 3.9, panel B). This is approximately 5.6 times higher than Mngoji (112.3 kg/ha), the lowest-scoring site in the entire study. The substantial biomass at Mnazi Bay reflects the long-term management history of this area and established no-take zones that have

allowed fish populations to recover. Msangamkuu (289.5 kg/ha) showed intermediate biomass above the depletion threshold, suggesting partial recovery despite lacking formal MMA status.

Table 3.6: Mean fish density (individuals per 100 m²) by family across the surveyed reefs in Mt-wara. Values represent the median density observed across all transects per site.

Family	Chamba Cha Kati	Nusura	Viulu
Pomacentridae	254.6	26.0	22.0
Scaridae	11.0	10.6	8.8
Labridae	7.4	11.0	8.2
Acanthuridae	5.0	3.6	3.4
Holocentridae	3.8	-	-
Chaetodontidae	3.6	1.0	1.6
Mullidae	1.6	3.6	5.0
Diodontidae	1.2	1.2	1.6
Balistidae	0.8	-	-
Siganidae	0.8	-	4.4
Fistularidae	0.4	-	-
Scorpaenidae	0.4	-	-
Lethrinidae	-	2.0	2.4
Lutjanidae/Nemipteri	-	1.2	1.2
Others	-	3.2	-
Penguipedidae/Synod	-	1.0	0.6
Pomacanthidae	-	1.8	2.8
Serranidae	-	0.8	0.4
Pempheridae	-	-	1.2

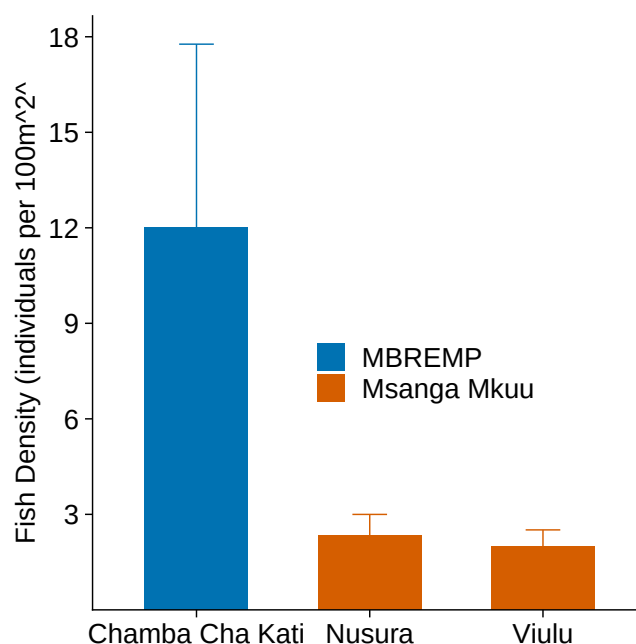


Figure 3.6: Median fish density (individuals per 100 m² +/- SE) across surveyed reefs in Mtwara. The dashed blue line represents the regional depletion threshold of 20 individuals per 100 m², indicating that most surveyed reefs currently sustain fish populations above critical depletion levels.

3.2.3 Coral Recruits

Recruit densities in Mtwara closely mirrored the substrate and biomass patterns. Mnazi Bay recorded the highest recruit density of all seven study sites (5.2 recruits m⁻²), indicating a functioning reproductive cycle and positive recovery trajectory (see Figure 3.10, panel B). Msangamkuu showed moderate density (2.1 m⁻²), while Mngoji had critically low recruit density (0.6 m⁻²). Combined with low coral cover and high algal dominance, the low recruit density at Mngoji signals a reef with limited capacity for natural recovery.

Table 3.7: Mean recruit density (individuals per 100 m²) by family across the surveyed reefs in Mtwara. Values represent the median density observed across all transects per site.

Family	Chamba Cha Kati	Nusura	Viulu
Acropora	500	200	100
merulina	500	-	-
Alveopora	300	100	400
Lobophyllia	300	100	100
Millepora	300	-	100
Porites(m)	300	650	450
Fungia	200	100	250

Gardineroseris	200	-	-
Pavona	200	200	300
Plerogyra	200	-	-
Pocillopora	200	100	100
Favia	100	150	400
Favites	100	200	300
Galaxea	100	200	200
Halomitra	100	-	-
Herpolitha	100	-	-
Leptoseris	100	100	100
Platygyra	100	100	100
Seriatopora	100	-	-
Synarea	100	300	200
Astreopora	-	100	400
Echinopora	-	100	300
Merullina	-	100	-
Pachyseris	-	100	-
Physogyra	-	200	-
Psammocoral	-	100	100
Tubipora	-	100	-
Acanthastrea	-	-	100
Hydnophora	-	-	100
Psamocoral	-	-	100
Stylophora	-	-	250

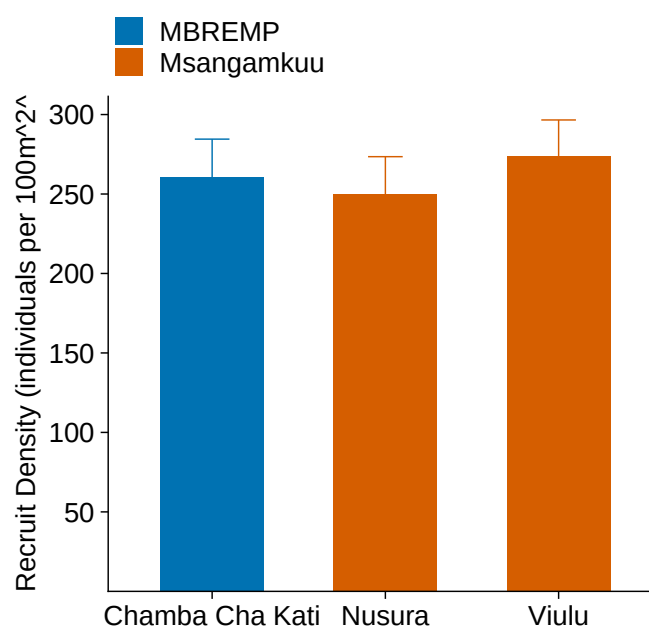


Figure 3.7: Median recruit density (individuals per 100 m² +/- SE) across surveyed reefs in Mtwara.

3.2.4 Invertebrate Assemblages

Indicator invertebrate densities in Mtwara varied significantly between sites and functional roles (Table 3.4). Msanga mkuu exhibited the highest densities of algal grazers (sea urchins) and nutrient cyclers (sea cucumbers), which are critical for maintaining reef health and supporting restoration efforts. In contrast, Mnazi Bay showed lower densities across most taxa, particularly for worms and Gastropods. Notably, Crown-of-Thorns starfish (COTs), a significant coral predator and threat to reef recovery, were recorded at Msanga mkuu, necessitating active monitoring to prevent outbreaks. The presence of (COTs) at Viuu reef (Figure 3.8) in Msangamkuu remains a major threat for restoration in Msanga mkuu.

Table 3.8: Mean recruit density (individuals per 100 m²) by family across the surveyed reefs in Mtwara. Values represent the median density observed across all transects per site.

Family	Chamba Cha Kati	Nusura	Viulu
Sea star	400	600	900
Urchin	400	500	1,150
Bivalves	300	350	500
Gastropods	250	400	800
Sea cucumber	150	300	200
Octopus	100	100	-
Worms	-	300	450

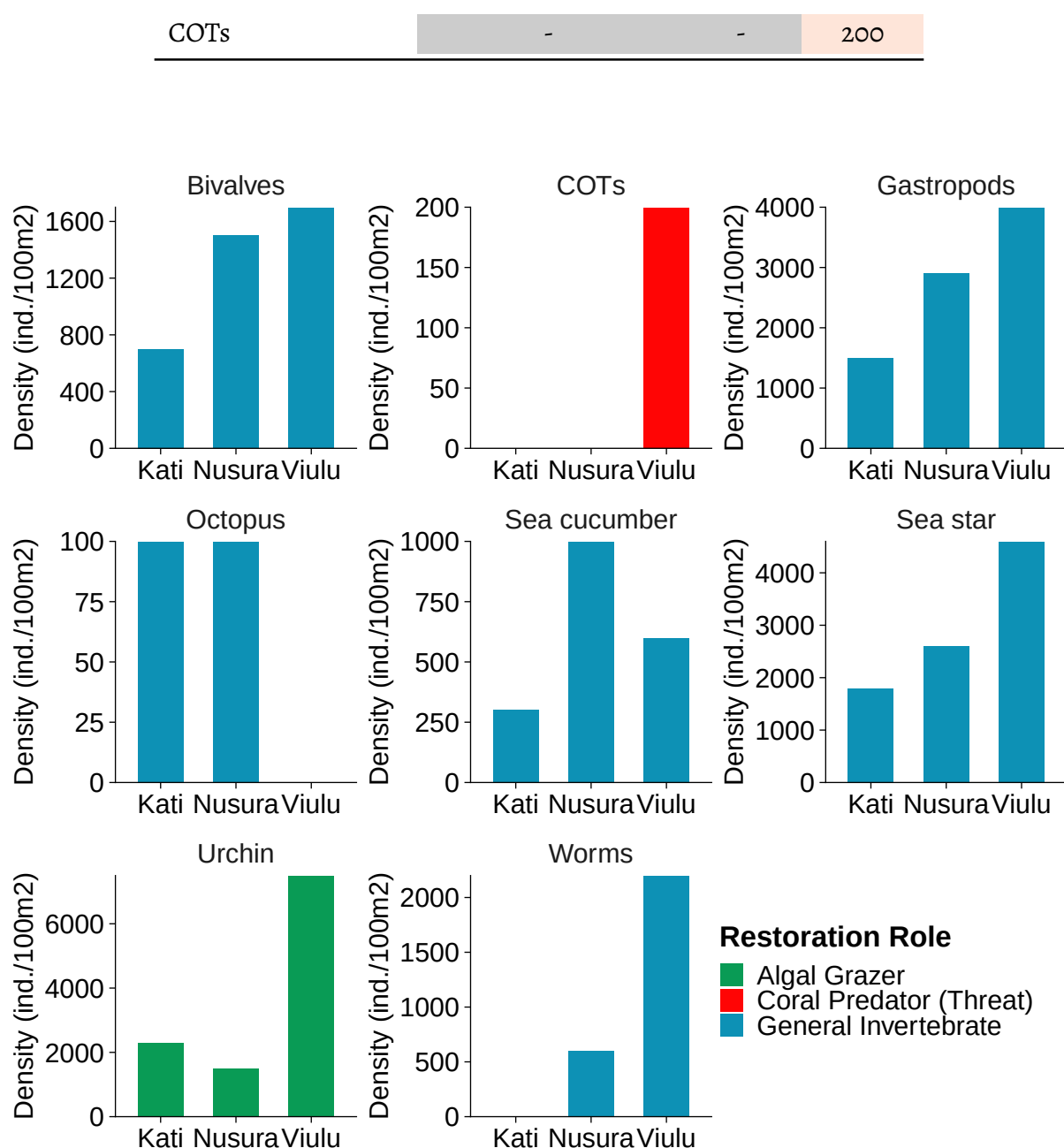


Figure 3.8: Invertebrate density (individuals per 100 m²) by taxon across reefs in Mnazi Bay and Msangamkuu.

3.3 Combined Regional Figures

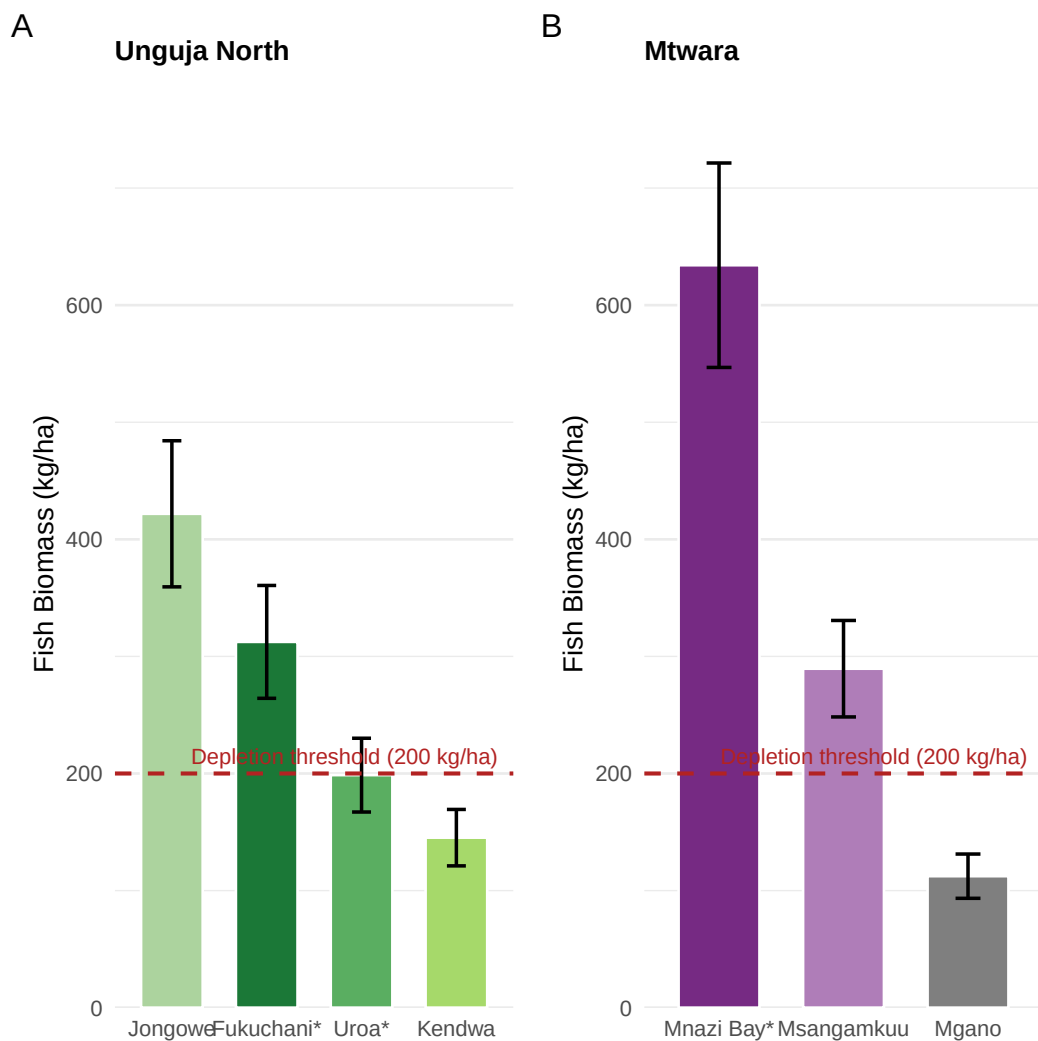


Figure 3.9: Fish biomass (kg/ha $\pm$ SE) at all study sites: (A) Unguja North and (B) Mtwara. Asterisks denote MMA sites. Dashed red line at 200 kg/ha is the commonly used depletion threshold for reef fish assemblages (McClanahan, Cinner, and Abunge 2014).

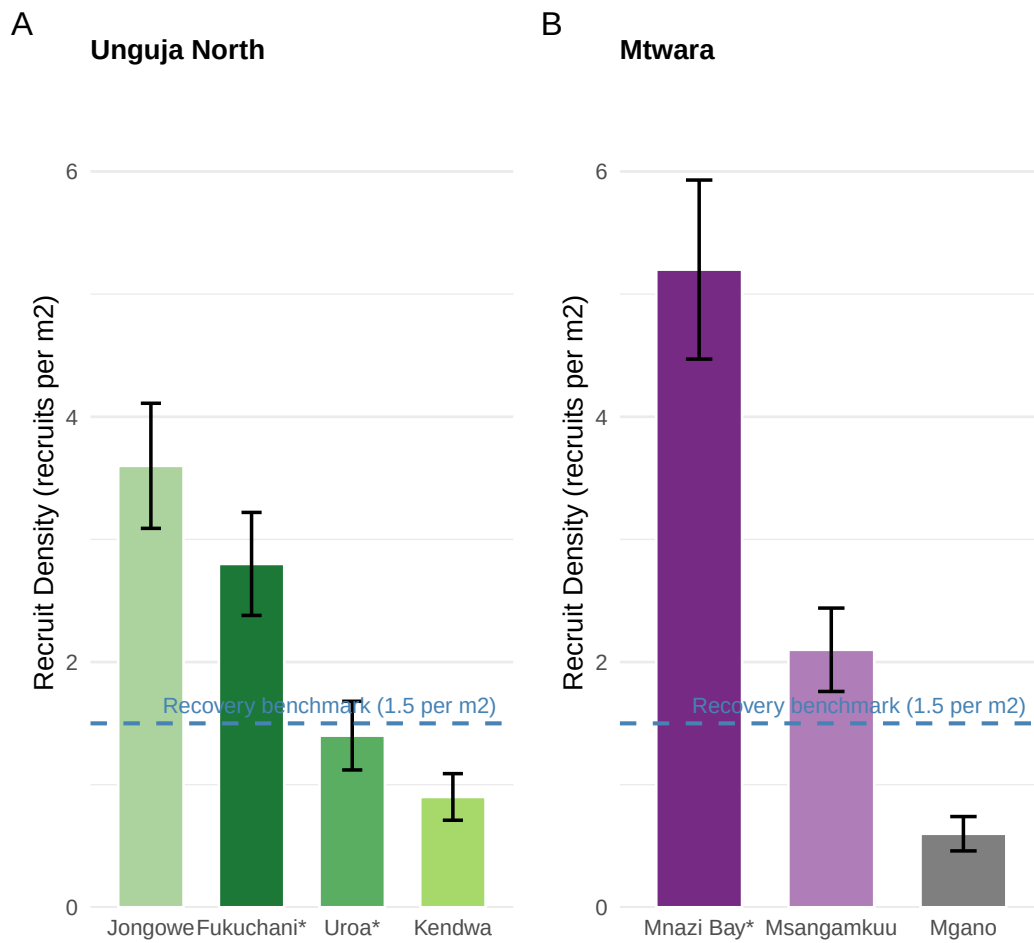


Figure 3.10: Coral recruit density (recruits m⁻² +/- SE) at all study sites: (A) Unguja North and (B) Mtwara. Asterisks denote MMA sites. Dashed blue line at 1.5 recruits m⁻² is the regional recovery benchmark.

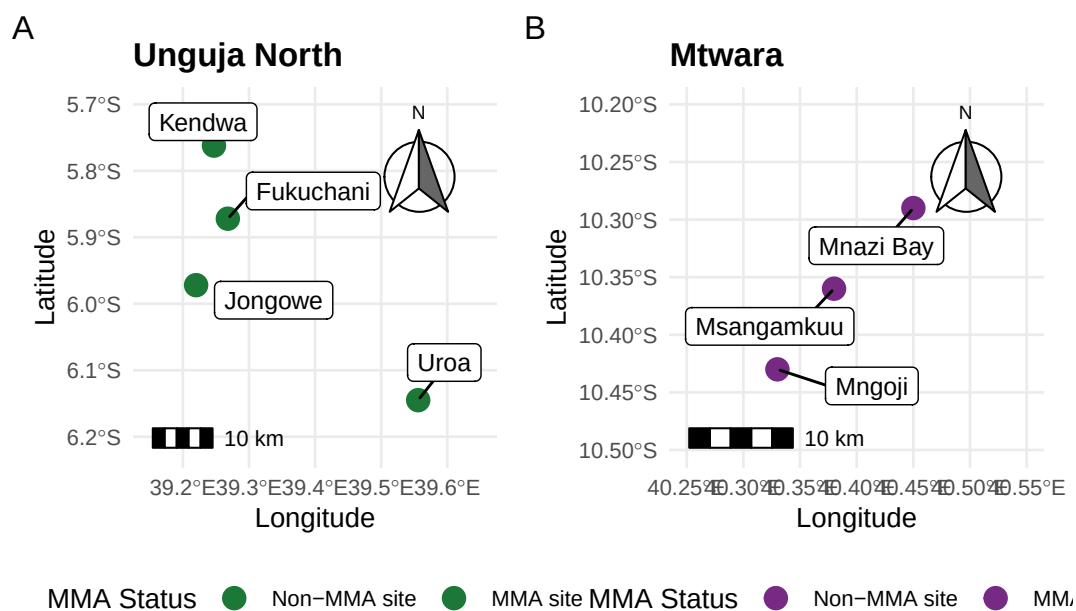


Figure 3.11: Map of study sites in (A) Unguja North, Zanzibar and (B) Mtwara, mainland Tanzania. Filled circles = MMA sites; open circles = non-MMA sites. Coordinates are approximate placeholders — replace with verified GPS waypoints from field data.

Note: Figure 3.11 uses approximate placeholder coordinates. Replace with verified GPS waypoints from field data collection.

4 Synthesis and Restoration Priorities

4.1 Ecological Condition Assessment

The conceptual framework in Figure 4.1 synthesises composite ecological condition scores (derived from the four assessed metrics) against governance and MMA effectiveness scores for all sites. This two-dimensional classification enables cross-regional comparison and restoration prioritisation.

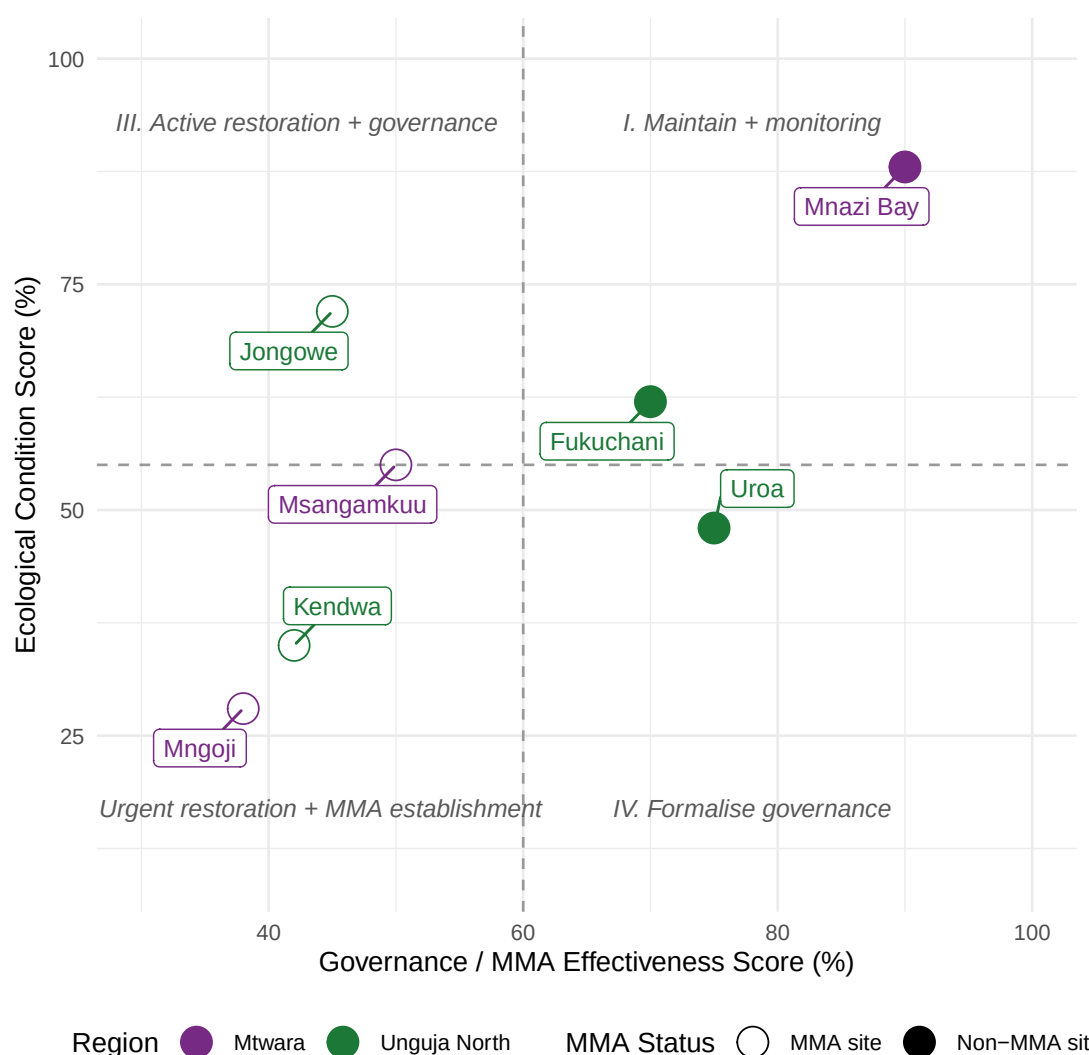


Figure 4.1: Conceptual restoration prioritisation framework. Sites are plotted by composite ecological condition score (y-axis; derived from benthic cover, fish biomass, recruit density, and invertebrate assemblages) against governance/MMA effectiveness score (x-axis). Dashed lines represent regional mean thresholds. Quadrant labels indicate recommended management actions.

Sites falling in the lower-left quadrant — **Mgano**, **Kendwa**, and to a lesser extent **Msanga Mkuu** — represent the highest restoration priority, combining poor ecological condition with weak or absent formal governance. **Uroa** falls near the condition threshold despite its MMA status, suggesting that management effectiveness requires strengthening at this site. **Mnazi Bay** performs best on both dimensions and should be treated as the regional reference site for restoration target-setting.

4.2 MMA Effectiveness

Across both regions, MMA sites recorded substantially higher fish biomass (mean: 456.1 kg/ha) and coral recruit density (mean: 3.5 recruits m^{-2}) compared with non-MMA sites (mean: 233.5 kg/ha and 1.7 recruits m^{-2} , respectively). These differences are consistent with WIO literature demonstrating that effectively managed no-take zones sustain significantly higher reef fish biomass than unprotected areas (Edgar et al. 2014; McClanahan, Cinner, and Abunge 2014; Cinner et al. 2016). Nonetheless, between-site heterogeneity — particularly the high biomass recorded at Jongowe (non-MMA) — underscores that MMA boundaries alone are insufficient predictors of reef condition, and that community-led resource management can drive positive ecological outcomes independently of formal protection status (Rocliffe et al. 2014).

4.3 Restoration Recommendations

Based on the assessment findings, a tiered restoration framework is recommended (see Table 4.1). Immediate interventions should focus on active coral gardening, macroalgae removal, and invertebrate restocking at the most degraded sites, paired with governance efforts to establish or strengthen MMA protections. Moderate-priority sites would benefit from targeted coral outplanting and enhanced surveillance, while maintenance-level monitoring is recommended for well-performing sites.

Table 4.1: Tiered restoration recommendations by site, prioritised by ecological condition and MMA governance status.

[!h]

Priority	Site	Region	Key Issues	Recommended Actions
Immediate Priority				
Immediate	Mgano	Mtwara	High algae; low coral, recruits, and invertebrates	Coral gardening, urchin and sea cucumber seeding, MMA establishment
Immediate	Kendwa	Unguja North	High algae; very low biomass, recruits, and invertebrates	Coral gardening, gear restrictions, MPA boundary extension
Moderate Priority				
Moderate	Uroa	Unguja North	Below-threshold biomass; moderate algae; sedimentation	Sediment management, coral outplanting, intensified MMA patrols
Moderate	Msanga Mkuu	Mtwara	Moderate condition; no formal MMA protection	Coral outplanting, MMA inclusion assessment, community surveillance
Maintenance				
Maintenance	Fukuchani	Unguja North	Moderate-good condition; MMA patrols need strengthening	Continue MMA patrols, community awareness, regular monitoring
Maintenance	Jongowe	Unguja North	Good condition but unprotected; informal governance present	Formalise community protection agreement, continue monitoring
Maintenance	Mnazi Bay	Mtwara	Best condition; long-term reference site preservation needed	Long-term monitoring; designate as restoration reference site

5Conclusions

This assessment provides the first systematic multi-metric evaluation of reef health across seven sites in Unguja North and Mtwara, Tanzania. Key findings are as follows:

1. **Mnazi Bay** (Mtwara) is the highest-performing site across all four ecological metrics and represents the closest analogue to a healthy reef reference state in this study.
2. **Mgano** and **Kendwa** are in critical condition, exhibiting low coral cover, high algal dominance, depleted fish biomass, and minimal recruit density. Both sites require immediate active restoration intervention.
3. MMA sites consistently outperform non-MMA sites across all four metrics, reinforcing the value of formalised marine governance. However, some non-MMA sites (e.g., Jongo) performed comparably to MMA sites, suggesting that community-led governance can supplement or substitute formal protection in certain contexts.
4. Indicator invertebrates — particularly sea cucumbers and giant clams — are severely depleted across nearly all sites, reflecting decades of targeted harvesting. Targeted restocking programmes and trade regulation should complement habitat restoration.
5. Future assessments should incorporate socioeconomic surveys to characterise community livelihood dependencies on each reef system and to design restoration programmes within a human-centred, rights-based framework.

References

- Babcock, RC et al. 1992. "Reproduction and Recruitment of Corals: Comparisons Among the Caribbean, the Tropical Pacific, and the Red Sea." *Marine Ecology Progress Series* 52: 109–25.
- Burke, Lauretta, K Reytar, M Spalding, and A Perry. 2011. *Reefs at Risk Revisited*. World Resources Institute.
- Cinner, Joshua E et al. 2016. "Bright Spots Among the World's Coral Reefs." *Nature* 535 (7612): 416–19.
- Edgar, Graham J et al. 2014. "Global Conservation Outcomes Depend on Marine Protected Areas with Five Key Features." *Nature* 506 (7487): 216–20.
- English, S, C Wilkinson, and V Baker. 1997. *Survey Manual for Tropical Marine Resources*. Australian Institute of Marine Science.
- Froese, Rainer, and Daniel Pauly. 2023. "FishBase." 2023. www.fishbase.org.
- Hodgson, G, J Hill, W Kiene, L Maun, J Mihaly, J Liebel, C Shauka, and R Devel. 2006. *Reef Check Instruction Manual: A Guide to Reef Check Coral Reef Monitoring*. Reef Check Foundation.
- McClanahan, Tim R et al. 2020. "Global Baselines and Benchmarks for Fish Biomass: Comparing Remote Reefs and Fisheries Closures." *Marine Ecology Progress Series* 641: 153–67.
- McClanahan, Tim R, Joshua E Cinner, and Caroline Abunge. 2014. "Perceived Benefits of Fisheries Management Restrictions in Madagascar." *Ecology and Society* 19 (1).
- Obura, David et al. 2016. "Coral Mortality Dynamics on Reefs of the Western Indian Ocean." *Coral Reefs*.
- Obura, David, Mishal Gudka, J Abdouravi, S Ahamada, et al. 2017. "Coral Reef Status Report for the Western Indian Ocean." *Global Coral Reef Monitoring Network (GCRMN)/International Coral Reef Initiative (ICRI)*.
- Pebesma, Edzer. 2018. "Simple Features for R: Standardized Support for Spatial Vector Data." *The R Journal* 10 (1): 439–46. <https://doi.org/10.32614/RJ-2018-009>.
- R Core Team. 2024. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Roccliffe, S, S Peabody, M Samoilys, and JP Hawkins. 2014. "Towards a Network of Locally Managed Marine Areas (LMMAs) in the Western Indian Ocean." *PLoS One* 9 (7): e103000.

- Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York. <https://ggplot2.tidyverse.org>.
- Wilkinson, Clive. 2008. *Status of Coral Reefs of the World: 2008*. Global Coral Reef Monitoring Network; Reef; Rainforest Research Centre.
- Wilson, JR et al. 2020. *Monitoring Coral Reefs: A Guide for the Western Indian Ocean*. CORDIO East Africa.
- Zhu, Hao. 2024. *kableExtra: Construct Complex Table with 'Kable' and Pipe Syntax*. <https://CRAN.R-project.org/package=kableExtra>.

Appendix

Appendix A: Full Data Tables

Appendix B: Survey Metadata

Table 5.1: Full benthic substrate composition data (% cover) for all seven sites.

[!h]

Region	Site	MMA	Hard Coral	Soft Coral	Algae	Rubble/Other
Unguja North						
Unguja North	Fukuchani	Yes	38.2	5.1	22.4	34.3
Unguja North	Uroa	Yes	29.7	4.8	31.5	34.0
Unguja North	Jongowe	No	42.1	6.3	18.9	32.7
Unguja North	Kendwa	No	24.5	3.9	38.7	32.9
Mtwara						
Mtwara	Mnazi Bay	Yes	51.3	7.2	15.4	26.1
Mtwara	Msanga Mkuu	No	35.8	5.5	27.3	31.4
Mtwara	Mgano	No	21.4	4.1	41.2	33.3

Table 5.2: Fish biomass (mean kg/ha plus or minus SE) for all seven sites.

[!h]

Region	Site	Biomass (kg/ha)	SE	MMA
Unguja North				
Unguja North	Fukuchani	312.4	48.2	Yes
Unguja North	Uroa	198.6	31.5	Yes
Unguja North	Jongowe	421.8	62.4	No
Unguja North	Kendwa	145.2	24.1	No
Mtwara				
Mtwara	Mnazi Bay	634.1	87.3	Yes
Mtwara	Msangamkuu	289.5	41.2	No
Mtwara	Mgano	112.3	18.9	No

Table 5.3: Coral recruit density (mean recruits m^{-2} plus or minus SE) for all seven sites.

[!h]

Region	Site	Density (m^{-2})	SE	MMA
Unguja North				
Unguja North	Fukuchani	2.8	0.42	Yes
Unguja North	Uroa	1.4	0.28	Yes
Unguja North	Jongowe	3.6	0.51	No
Unguja North	Kendwa	0.9	0.19	No
Mtwara				
Mtwara	Mnazi Bay	5.2	0.73	Yes
Mtwara	Msangamkuu	2.1	0.34	No
Mtwara	Mgano	0.6	0.14	No

Table 5.4: Indicator invertebrate density (individuals per 100 m²) by taxon and site for all seven sites.

[!h]

Region	Site	Taxon	Density (100 m ⁻²)	MMA
Unguja North				
Unguja North	Fukuchani	Sea Urchins	12.4	Yes
Unguja North	Fukuchani	Sea Cucumbers	3.2	Yes
Unguja North	Fukuchani	Giant Clams	1.8	Yes
Unguja North	Uroa	Sea Urchins	9.1	Yes
Unguja North	Uroa	Sea Cucumbers	2.7	Yes
Unguja North	Uroa	Giant Clams	0.9	Yes
Unguja North	Jongowe	Sea Urchins	15.3	No
Unguja North	Jongowe	Sea Cucumbers	4.1	No
Unguja North	Jongowe	Giant Clams	2.4	No
Unguja North	Kendwa	Sea Urchins	6.8	No
Unguja North	Kendwa	Sea Cucumbers	1.2	No
Unguja North	Kendwa	Giant Clams	0.4	No
Mtwara				
Mtwara	Mnazi Bay	Sea Urchins	18.9	Yes
Mtwara	Mnazi Bay	Sea Cucumbers	6.8	Yes
Mtwara	Mnazi Bay	Giant Clams	3.7	Yes
Mtwara	Msangamkuu	Sea Urchins	11.2	No
Mtwara	Msangamkuu	Sea Cucumbers	3.4	No
Mtwara	Msangamkuu	Giant Clams	1.5	No
Mtwara	Mgano	Sea Urchins	5.4	No
Mtwara	Mgano	Sea Cucumbers	0.8	No
Mtwara	Mgano	Giant Clams	0.2	No

Table 5.5: Survey metadata for all seven study sites. Replace placeholder values with actual field data.

[!h]

Site	Region	Survey Date	Depth Range (m)	Transects (n)	Surveyor(s)	Visibility (m)
Unguja North						
Fukuchani	Unguja North	[Insert date]	3-12	6	[Insert name(s)]	[Insert]
Uroa	Unguja North	[Insert date]	4-11	6	[Insert name(s)]	[Insert]
Jongowe	Unguja North	[Insert date]	3-12	6	[Insert name(s)]	[Insert]
Kendwa	Unguja North	[Insert date]	3-10	6	[Insert name(s)]	[Insert]
Mtwara						
Mnazi Bay	Mtwara	[Insert date]	5-14	6	[Insert name(s)]	[Insert]
Msanga Mkuu	Mtwara	[Insert date]	4-12	6	[Insert name(s)]	[Insert]
Mgano	Mtwara	[Insert date]	3-11	6	[Insert name(s)]	[Insert]

Index

- appendix
 - raw data tables, [28](#)
 - survey metadata, [28](#)
- benthic substrate
 - Mtwara, [12](#)
 - point-intercept transect, [5](#)
 - Unguja North, [7](#)
- coral recruits
 - juvenile density, [5](#)
 - Mtwara, [15](#)
 - Unguja North, [9](#)
- coral reef
 - ecology, [1](#)
- fish biomass
 - belt transect, [5](#)
 - Mtwara, [13](#)
 - Unguja North, [8](#)
- GPS coordinates, [4](#)
- invertebrates
 - Mtwara, [17](#)
 - survey, [6](#)
 - Unguja North, [11](#)
- marine managed areas (MMAs), [1](#)
 - effectiveness, [23](#)
- methodology
 - field survey, [2](#)
- Mtwara, [2](#)
 - results, [12](#)
- reef restoration, [1](#)
- restoration
 - priorities, [22](#)
 - recommendations, [23](#)
- Tanzania
 - coral reefs, [1](#)
- Unguja North, [2](#)
 - results, [7](#)